12.18 Memorylessness

WORKSHEET

The question

STEP III 2021, Question 11

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself What is $\mathrm{P}(X > x)$ for $x \geq 0$? What is $\mathrm{P}(X > s + t \mid X > s)$?

STAGE 1 The whole part

Covers: Part (i)

Ask yourself $Y = n$ exactly when $n \leq X < n + 1$. What is $\mathrm{P}(X > x)$?

STAGE 2 The fractional part

Covers: Part (ii)

Pause and try $Z < z$ means X lies in one of the intervals $[n, n+z)$. Add up their probabilities: you get a geometric series.

STAGE 3 The mean of Z

Covers: Part (iii)

Ask yourself Either differentiate Part (ii) to get the density of Z and integrate z times it, or use $\mathrm{E}(Z) = \int_0^1 \mathrm{P}(Z > z)\,\mathrm{d}z$.

STAGE 4 Independence

Covers: Part (iv)

Ask yourself Write $\mathrm{P}(Y = n \text{ and } z_1 < Z < z_2)$ as a probability about X. Compare with the product of your answers to Parts (i) and (ii).

What's really going on

Why the exponential is special

The independence of Y and Z comes straight from $e^{-\lambda(n+z)} = e^{-\lambda n}e^{-\lambda z}$, and that is the memoryless property in disguise. In fact the exponential is the *only* continuous distribution without memory: if $\mathrm{P}(X > s+t) = \mathrm{P}(X > s)\mathrm{P}(X > t)$ for all $s, t \geq 0$, then $g(t) = \ln \mathrm{P}(X > t)$ satisfies $g(s+t) = g(s) + g(t)$, and (for a monotonic g) this forces $g(t) = -\lambda t$. That is why the exponential models waiting times for events that happen "at random", with no build-up: radioactive decays, the arrival of customers, the failure of electronic components that do not wear out.

Minima and the waiting-time paradox

If X_1 and X_2 are independent exponentials with rates λ_1 and λ_2, then $\mathrm{P}(\min > x) = e^{-\lambda_1 x}e^{-\lambda_2 x}$, so the minimum is exponential with rate $\lambda_1 + \lambda_2$. Waiting for the first of two independent random events is like waiting for one event at the combined rate (STEP III 2002 Q13, in the exercises). Memorylessness also explains the *waiting-time paradox*: if buses arrive at random with an average gap of 10 minutes, then a passenger arriving at a random time waits on average 10 minutes, not 5, because the remaining wait is memoryless; and the gap they arrive in is on average 20 minutes long, since long gaps are more likely to be landed in.

Rutherford and the Poisson process

In 1910 Ernest Rutherford and Hans Geiger counted the alpha particles emitted by a radioactive source in intervals of 7.5 seconds, and found that the counts followed the Poisson distribution, as the mathematician Harry Bateman showed they should if the emissions occurred independently at a constant rate. The times between emissions are

then exponential. The *Poisson process*, a stream of events with exponential gaps, is now the basic model of randomness in time, used for telephone calls, insurance claims and the firing of neurons; the next section, Section 12.19, extends it to space.

Try this next

1. **Two bus routes.** Try STEP III 2002, Question 13 (step.maths.org/questions/02-s3-q13). It uses the minimum of two exponential random variables to find the wait for the next bus.

2. **Memoryless discrete.** Show that the geometric distribution is memoryless: $\mathrm{P}(Y \geq m+n \mid Y \geq m) = \mathrm{P}(Y \geq n)$.

3. **The waiting-time paradox.** Buses arrive as a Poisson process with rate λ. Show that the gap containing a fixed time t (for large t) has expected length $\frac{2}{\lambda}$, twice the average gap.

4. **The density of Z.** Find the probability density of Z, and sketch it for $\lambda = 1$ and $\lambda = 5$.

Further reading

- **Geoffrey Grimmett & Dominic Welsh, *Probability: An Introduction*** (Oxford University Press), on the exponential distribution and the Poisson process.
- **Sheldon Ross, *Introduction to Probability Models*** (Academic Press). Exponential distributions, Poisson processes and queues.
- **David Spiegelhalter, *The Art of Uncertainty*** (Pelican). Randomness and waiting in everyday life.

The full guided solution is in Section 12.18:
https://beyondstep.keithmcnulty.org/ch12/memoryless.html